DATAX500

1631819

2026-03-22

Github Repository
Moodle Submission

Assignment 1

1. Using the '*heart.csv*' data-set, calculate the following statistical measures for age in *R* **(6 marks)**

- First, we load the data-set:

```
# load automatically via git files
heart<-read.csv("Heart.csv")


## Alternatively, load the dataset manually using:
# heart<-read.csv(file.choose())
```

a) Mean **(2 marks)**

```
# assign age column to "age" var for easier processing
age<-heart$age

# calculate the mean age
mean_age<-mean(age)
mean_age
```

```
## [1] 54.01038
```

```
# round to 2 d.p.
mean_age<-round(mean_age, 2)
mean_age
```

```
## [1] 54.01
```

b) Median **(2 marks)**

```r
# calculate the median age
median_age<-median(age)
median_age
```

```
## [1] 54
```

c) Sample standard deviation **(2 marks)**

```r
# calculate the sample standard deviation
sd_age<-sd(age)
sd_age
```

```
## [1] 9.132316
```

```r
# round sample standard deviation to 2 d.p.
sd_age<-round(sd_age, 2)
sd_age
```

```
## [1] 9.13
```

---

2. A researcher wishes to investigate the relationship between gender and cholesterol levels.

   Key-points to notice here is that **we are measuring cholesterol levels** based on gender, making **cholesterol** the *dependent variable*, and **gender** the *independent variable*.

   Using the dataset, perform an *appropriate statistical test* in *R* and interpret the output. Your answer should include a **null hypothesis**, an **alternative hypothesis** , and a **p-value** **(5 marks)**

   - Since *gender* (categorical) and *cholesterol* (continuous) are **independent** from eachother, we conduct an independent (two-sampled) t-test to test the following hypotheses:

     Null hypothesis: H0: U1 = U2

     - There is no difference in mean cholesterol levels between genders.

     Alternative hypothesis: H1: U1 != U2

     - There is a significant difference in mean cholesterol levels between genders.

     p-value:

     ```r
     # peek at the dataset so we can view the column names
     head(heart, n=2)
     ```

     ```
     ##   age gender cp trtbps chol fbs restecg maxheartrate exng
     ## 1  60      1  3    145  233   1       0          150    0
     ## 2  35      1  2    130  250   0       1          187    0
     ```

     ```r
     # perform a t-test on the heart data set between chol and gender
     t.test(chol~gender, data=heart)
     ```

     ```
     ##
     ##  Welch Two Sample t-test
     ##
     ## data:  chol by gender
     ```

```
## t = 2.9765, df = 133.12, p-value = 0.003464
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to
## 95 percent confidence interval:
##   7.289758 36.169316
## sample estimates:
## mean in group 0 mean in group 1
##        262.6989        240.9694
```

Interpretation:

- The Welch two-sample t-test provides **evidence to reject the null hypothesis**, that **there will be no difference in mean cholesterol levels between genders, p=0.003**.
- This interpretation is further supported with **a large t-value of 2.98.**
- **Further testing is recommended** to determine the source or magnitude of this cholesterol difference.

---

3. A researcher wishes to investigate the predictors of maximum heart rate. Using the dataset provided, fit an apprpopriate statistical model of your choice. Make sure to include age as one of the predictors **(14 marks)**

   Since *maximum heart rate* and *age* are both continuous variables, a regression model will probably be suitable for this analysis where we test for a relationship between ones age and their maximum heart rate.

   a) What are the null and alternative (two-sided) hypotheses associated with age? **(2 marks)**

   Null hypothesis: H0: B1 = 0

   - The regression coefficient for age is zero (i.e., age has no linear effect on maximum heart rate.) **Note:** This means that for every x increase in age, we should not observe a fixed change in y for heart rate.

   Alternative hypothesis: H1: B1 != 0

   - There will be a significant relationship between age and maximum heart-rate.

   b) What is the estimated coefficient of age and what does this imply in terms of the relationship between age and maximum heart rate? **(3 marks)**

```r
# again, peek at the dataset so we can see what we're working with
head(heart, n=2)
```

```
##   age gender cp trtbps chol fbs restecg maxheartrate exng
## 1  60      1  3    145  233   1       0          150    0
## 2  35      1  2    130  250   0       1          187    0
```

```r
# store the mean maxheartrate in a variable
rate<-heart$maxheartrate

# create a linear model using maxheartrate and age
model<-lm(maxheartrate ~ age, data=heart)
# ouput the model summary for analysis
summary(model)
```

```
##
## Call:
## lm(formula = maxheartrate ~ age, data = heart)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -66.297 -11.263   3.842  15.695  44.758
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 204.0145     7.4408   27.42  < 2e-16 ***
## age          -0.9958     0.1358   -7.33 2.35e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 21.05 on 287 degrees of freedom
## Multiple R-squared:  0.1577,   Adjusted R-squared:  0.1548
## F-statistic: 53.73 on 1 and 287 DF,  p-value: 2.347e-12
```

c) Comment on the predictors you have included in the model and justify the inclusion of these variables in your model **(3 marks)**

d) interpret the model output in plain English. Your interpretation must include the p-value associated with each coefficient (slopes and intercept) test, and R-squared value **(3 marks)**

e) Comment on whether the assumptions of a linear regression model have been met. Include appropriate statistical plots to support your answer **(3 marks)**

Data dictionary:

cp: chest pain type

1. typical angina
2. atypical angina
3. non-anginal pain
4. asymptomatic

trtbps:

- resting blood pressure (in mm Hg)

chol:

- cholesterol (in mg/dl)

gender:

1. female;
2. male;

fbs: (fasting blood sugar > 120 mg/dl)

0. False
1. True

rest_ecg: resting electrocardiographic result

0. Normal
1. Having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mV)
2. Showing probably or definite left ventricular hypertrophy by Estes' criteria

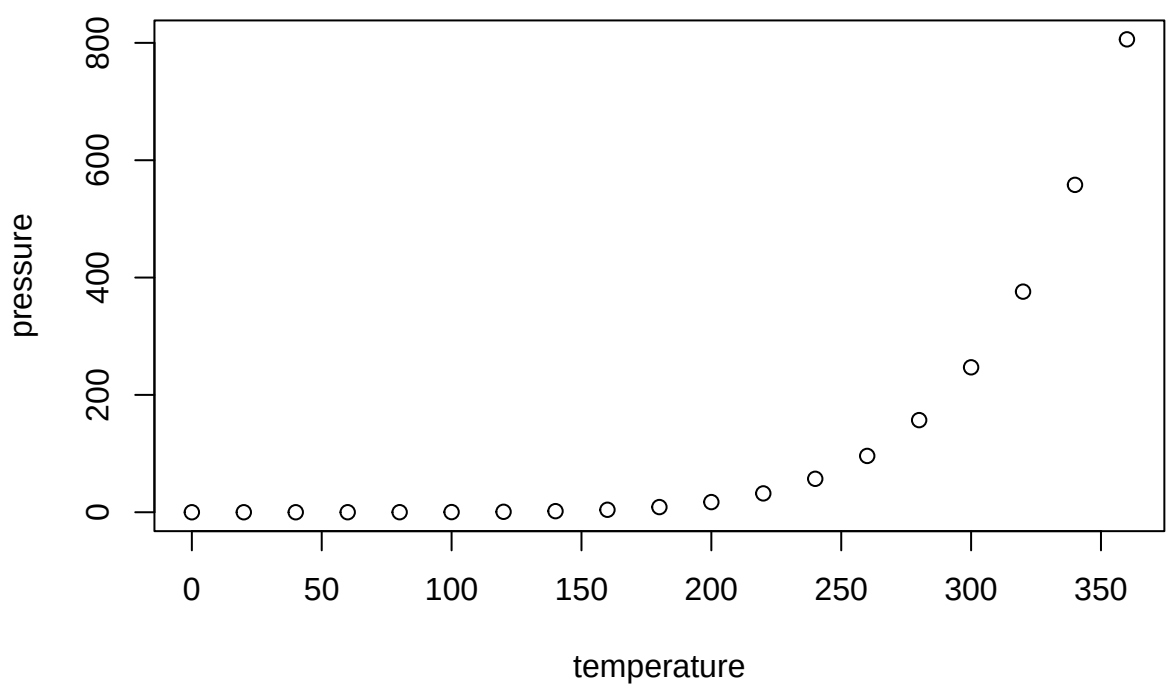
---

```r
summary(cars)
```

```
##      speed           dist       
##  Min.   : 4.0   Min.   :  2.00  
##  1st Qu.:12.0   1st Qu.: 26.00  
##  Median :15.0   Median : 36.00  
##  Mean   :15.4   Mean   : 42.98  
##  3rd Qu.:19.0   3rd Qu.: 56.00  
##  Max.   :25.0   Max.   :120.00  
```

Including Plots

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.